

TOPIC 02 • TEXT PREPROCESSING 1

Tokenization and Basic Terminologies

The four words we will use in every NLP lesson, and the first step of cleaning text.

Four Terms to Know

01

Corpus = paragraph

The whole body of text we are working with.

02

Documents = sentences

Each sentence inside the corpus.

03

Vocabulary = unique words

Every distinct word, counted once. Like a dictionary.

04

Words = all words

Every word in the corpus, repeats included.

TOKENIZATION

Paragraph → Sentence Tokens

CORPUS

"My name is Riya and I love learning Python, ML and NLP. I also write a tech blog."

↓ Tokenize into sentences, split at . or !

Token 1 · document 1

My name is Riya and I love learning Python, ML and NLP

Token 2 · document 2

I also write a tech blog

TOKENIZATION, ONE LEVEL DOWN

Sentence → Word Tokens

Document 1

My name is Riya and I love learning Python, ML and NLP

↓ Tokenize into words

My name is Riya and I love learning Python ML and NLP

A token can be a **sentence** or a **word**.

Why: every word must later become a **vector**.

WORKED EXAMPLE

Words vs Vocabulary

CORPUS "I like to read story books. My sister likes comic books."

Document 1

I like to read story books

Document 2

My sister likes comic books

Total words

11

Unique words = **vocabulary**

10

books appears twice, so it is counted once in the vocabulary.

“Like” and “Likes” Are Different Words

BEFORE

I **like** to read story books. My sister **likes** comic books.

Words **11** Vocabulary **10**



AFTER

I **like** to read story books. My sister **like** comic books.

Words **11** Vocabulary **9**

To a computer, **like** and **likes** are unrelated. Lemmatization will fix this later.

What Is Tokenization?

Tokenization is the process of converting a **paragraph** or a **sentence** into **tokens**.

Paragraph → **sentences**

each sentence is a token

Paragraph or sentence → **words**

each word is a token

→ Next: tokenization in Python with NLTK.